

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

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RESEARCH CONTRIBUTION

More AI reviewers don't help — **decorrelation does**. Agreement predicts truth only when the reviewers are independent.

89%

3 independent families agree → matches ground truth

54%

one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

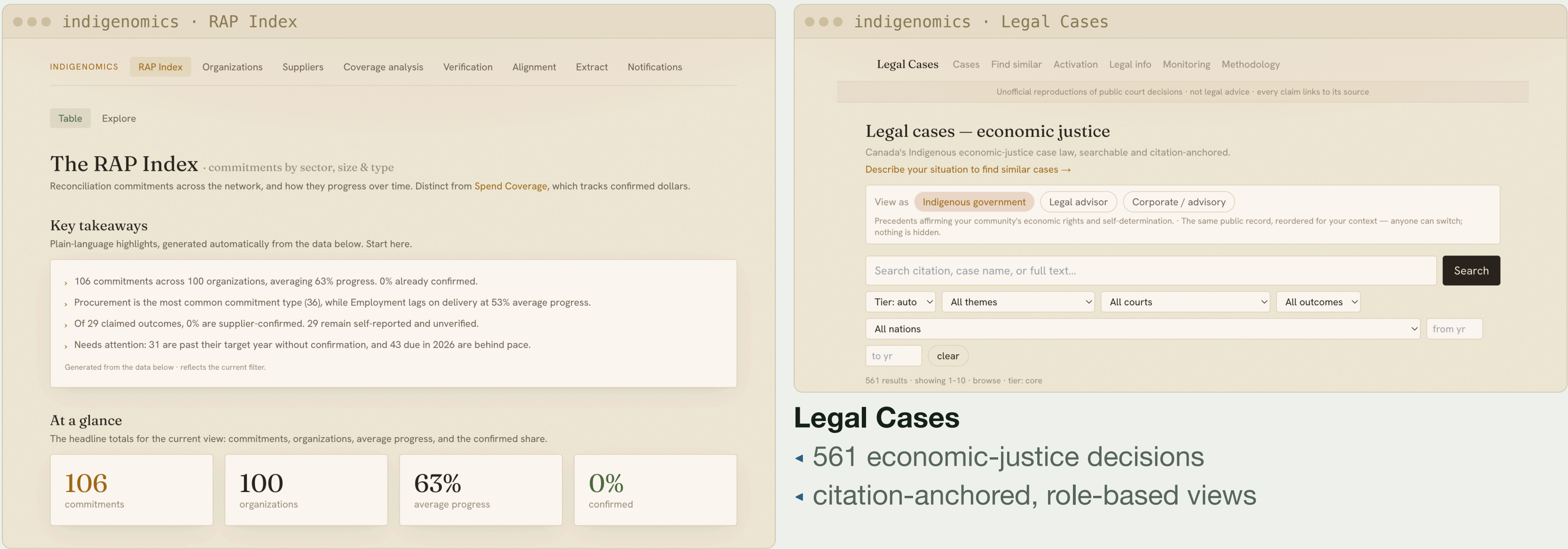
Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.

OUR SOLUTION

▪Searchable RAP commitments ▪Legal precedent database ▪AI extraction from PDFs
▪Human review workflow ▪AWS serverless hosting ▪Authentication & roles ▪Progress tracking
▪Coverage analytics

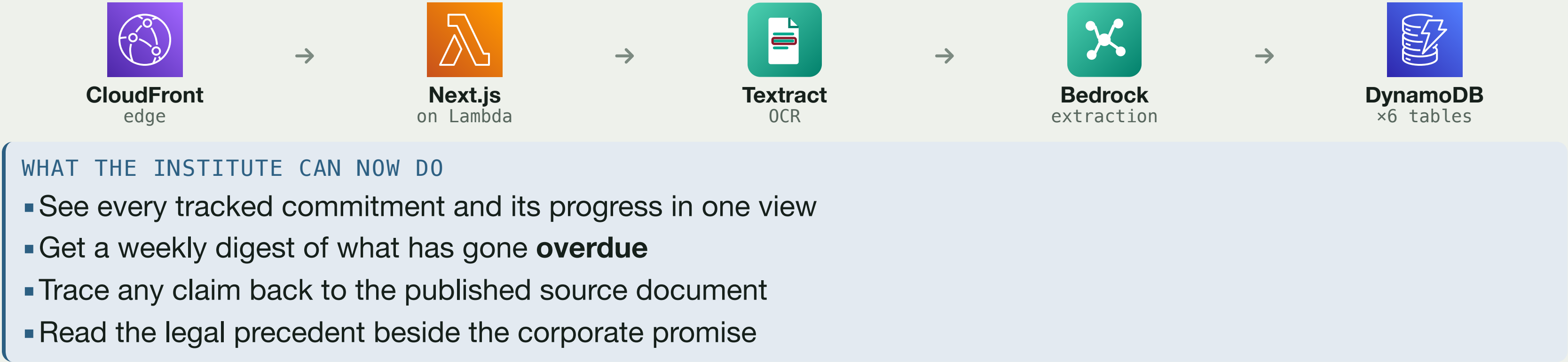
WHO USES IT

▪Indigenomics Institute ▪Indigenous organizations ▪Corporate partners ▪Researchers & legal advisors



RAP Index
• 106 commitments · 100 organizations
• 63% average progress tracked

ARCHITECTURE — SERVERLESS ON AWS



PROJECT IMPACT

▪Production platform delivered to the Indigenomics Institute
▪Reconciliation commitments made searchable and verifiable
▪Indigenous legal precedent explorer, citation-anchored

02 HOW IT BECAME RESEARCH

Client problem

reconciliation data scattered & unverifiable

We built the portal

production platform, shipped on AWS

179 pull requests

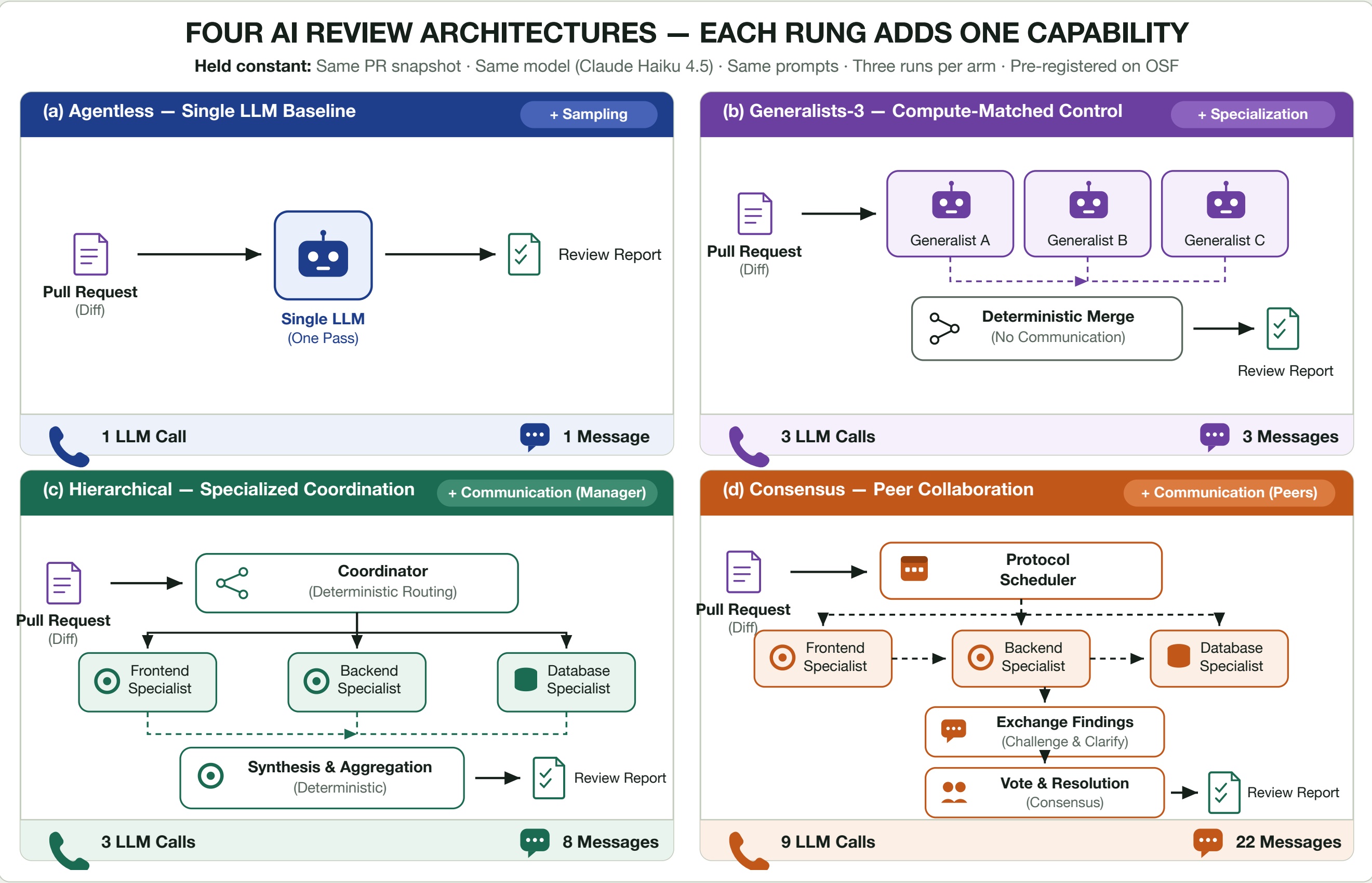
30 of them ours — real, merged, AI-reviewed

Benchmark dataset

E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews

Four AI review architectures

each rung adds exactly one capability

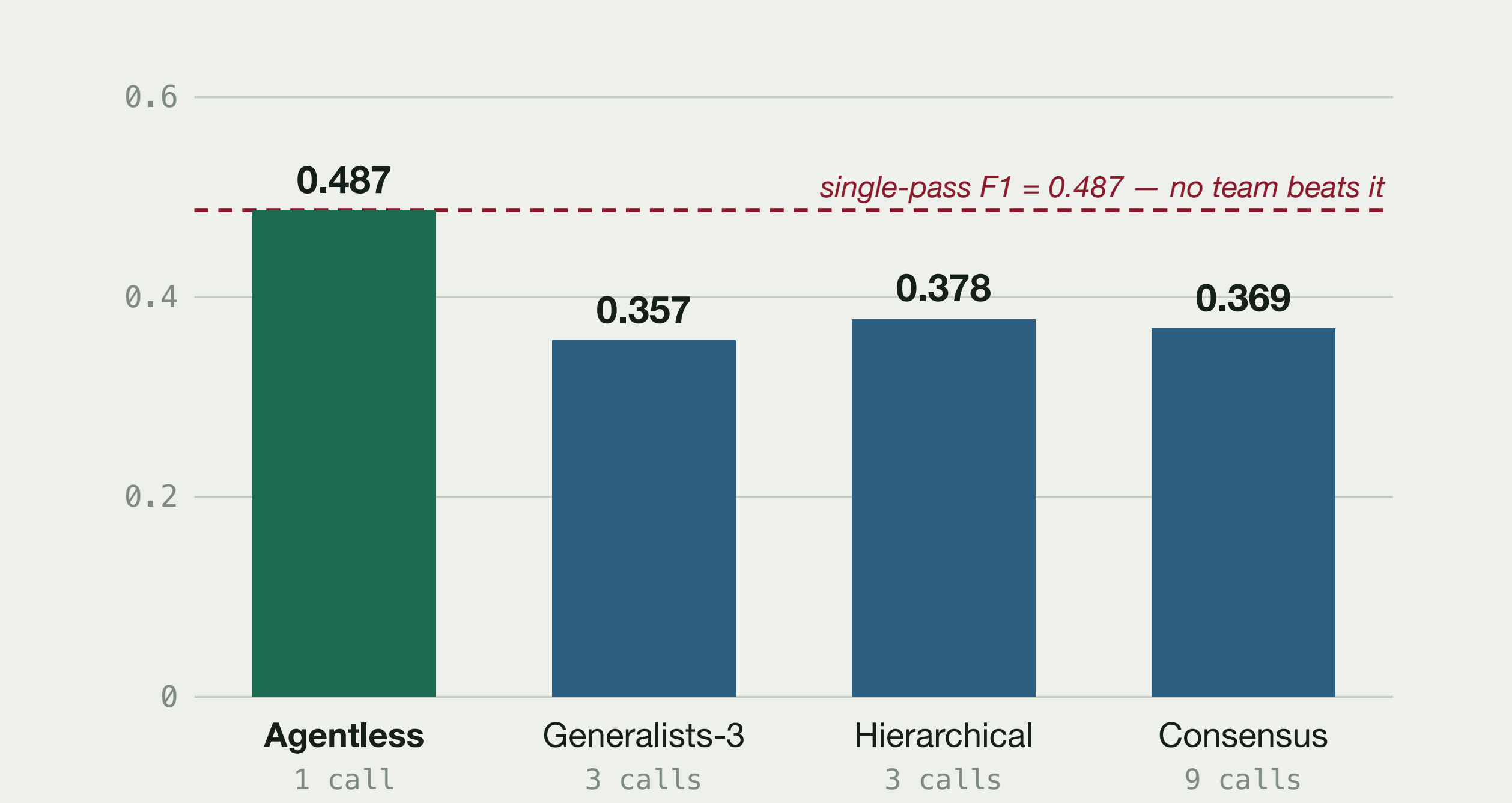


RESEARCH IMPACT

▪First compute-matched comparison of review architectures
▪Independence beats more agents — decorrelate your verifiers
▪Guidance for anyone deploying AI code review in production

03 WHAT WE FOUND · E1 · QODO, 99 PRS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling ($p=0.73$); peer debate adds nothing over a manager ($p=0.12$).



Extra agents buy recall at a price — up to **5x** the cost per confirmed finding.



Bounded. Recall plateaus at **≈0.83**; on our own unlabeled code, LLM judges disagree ($\kappa=0.03$).

WHERE THE SIGNAL LIVES

Agreement recovers **universal functional bugs** far better than project-specific conventions — which is why the ceiling exists.

43%

functional-bug recall at 3 independent families agreeing

18%

rule-violation recall at the same depth

Pair decorrelated review with a deterministic linter.



Code & data
rap-review-research



Live portal
try the demo